

ASSIGNMENT 21.2 AI-ASSISTED-CODING

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TASK 1: AI-ASSISTED PAIR PROGRAMMING

INITIAL PROGRAM GENERATED WITH AI HELP -
SIMPLE TO-DO LIST MANAGER.

```
# TODO.PY - INITIAL  
VERSION FROM AI TODOS =  
[]  
  
DEF ADD_TASK(TASK):  
    TODOS.APPEND(TASK)  
    PRINT(F"ADDED: {TASK}")  
  
DEF VIEW_TASKS():  
    FOR  
    I, TASK IN  
    ENUMERATE(TODOS, 1):  
        PRINT(F"{I}. {TASK}")  
  
DEF  
REMOVE_TASK(INDEX):  
    IF 0 < INDEX <=  
    LEN(TODOS):
```

```

    REMOVED =
    TODOS.POP(INDEX - 1)
    PRINT(F"REMOVED:
    {REMOVED}")    ELSE:
        PRINT("INVALID INDEX")

    WHILE TRUE:
        PRINT("\n1. ADD 2. VIEW 3.
        REMOVE 4. QUIT")    CHOICE =
        INPUT("CHOOSE: ")    IF CHOICE
        == '1':        TASK =
        INPUT("TASK: ")
        ADD_TASK(TASK)    ELIF
        CHOICE == '2':
        VIEW_TASKS()    ELIF CHOICE
        == '3':        VIEW_TASKS()

```

```
        IDX = INT(INPUT("INDEX  
TO REMOVE: "))  
REMOVE_TASK(IDX)    ELIF  
CHOICE == '4':  
    BREAK
```

AI SUGGESTED ENHANCEMENTS: ADD ERROR HANDLING, SAVE/LOAD FROM FILE, MARK COMPLETE.

IMPROVED VERSION AFTER AI SUGGESTIONS:


```
# TODO_IMPROVED.PY - ENHANCED WITH AI HELP:  
ERROR HANDLING, PERSISTENCE, COMPLETE  
FEATURE IMPORT JSON IMPORT OS
```

```
TODO_FILE = 'TODOS.JSON'
```

```
DEF LOAD_TODOS():    IF  
OS.PATH.EXISTS(TODO_FILE):  
WITH OPEN(TODO_FILE, 'R')  
AS F:  
    RETURN  
JSON.LOAD(F)    RETURN []
```

```
DEF SAVE_TODOS(TODOS):  
WITH OPEN(TODO_FILE, 'W')  
AS F:  
    JSON.DUMP(TODOS, F)
```

```
TODOS = LOAD_TODOS()

DEF ADD_TASK(TASK):
    TODOS.APPEND({"TASK": TASK,
"COMPLETED": FALSE})
    SAVE_TODOS(TODOS)
    PRINT(F"ADDED: {TASK}")

DEF VIEW_TASKS():
    IF NOT TODOS:
        PRINT("NO TASKS.")
    RETURN    FOR I, ITEM IN
    ENUMERATE(TODOS, 1):
        STATUS = "✓" IF
    ITEM["COMPLETED"] ELSE " "
    PRINT(F"{I}. [{STATUS}]
{ITEM['TASK']}")
```

```

DEF
REMOVE_TASK(INDEX):
TRY:      IF 0 < INDEX <=
LEN(TODOS):

        REMOVED =
TODOS.POP(INDEX - 1)
SAVE_TODOS(TODOS)
PRINT(F"REMOVED:
{REMOVED['TASK']}")      ELSE:

        PRINT("INVALID
INDEX")    EXCEPT
VALUEERROR:

        PRINT("ENTER VALID NUMBER")

DEF
COMPLETE_TASK(INDEX):
TRY:

```



```

        IF 0 < INDEX <= LEN(TODOS):
            TODOS[INDEX -
1][ "COMPLETED"] = TRUE
SAVE_TODOS(TODOS)
PRINT("MARKED COMPLETE")
ELSE:
    PRINT("INVALID
INDEX")    EXCEPT
VALUEERROR:
    PRINT("ENTER VALID NUMBER")

WHILE TRUE:
    PRINT("\n1. ADD 2. VIEW 3. REMOVE
4. COMPLETE 5. QUIT")    CHOICE =
INPUT("CHOOSE: ")    IF CHOICE == '1':
        TASK = INPUT("TASK:
").STRIP()    IF TASK:

```

```
ADD_TASK(TASK)
ELSE:
    PRINT("TASK CANNOT
BE EMPTY")  ELIF CHOICE ==
'2':    VIEW_TASKS()  ELIF
CHOICE == '3':
VIEW_TASKS()    IDX =
INT(INPUT("INDEX: "))
REMOVE_TASK(IDX)  ELIF
CHOICE == '4':
VIEW_TASKS()    IDX =
INT(INPUT("INDEX: "))
COMPLETE_TASK(IDX)  ELIF
CHOICE == '5':
    BREAK  ELSE:
    PRINT("INVALID CHOICE")
```

COMPARISON: ORIGINAL WAS BASIC CONSOLE APP WITHOUT PERSISTENCE OR COMPLETION. AI ADDED JSON SAVE/LOAD, COMPLETION STATUS, INPUT VALIDATION - MAKES IT PRODUCTION-READY.

TASK 2: VERSION CONTROL INTEGRATION
INITIALIZED GIT REPO FOR THE PROJECT.

```
$ MKDIR TODO-AI-LAB
```

```
$ CD TODO-AI-LAB
```

```
$ GIT INIT
```

```
INITIALIZED EMPTY GIT REPOSITORY IN  
/HOME/USER/TOD0-AI-LAB/.GIT/
```

```
$ GIT BRANCH -M MAIN
```

```
$ GIT ADD TODO.PY
```

```
$ GIT COMMIT -M "INITIAL AI-GENERATED TO-DO  
LIST MANAGER"
```

[MAIN (ROOT-COMMIT) ABC1234] INITIAL AI-
GENERATED TO-DO LIST MANAGER

1 FILE

CHANGED, 25

INSERTIONS(+)

CREATE MODE

100644

TODO.PY

CREATED NEW BRANCH FOR IMPROVEMENTS.

\$ GIT CHECKOUT -B AI-ENHANCEMENTS
SWITCHED TO A NEW BRANCH 'AI-
ENHANCEMENTS'

COPIED IMPROVED CODE TO TODO.PY,
ADDED IMPROVEMENTS

\$ GIT ADD TODO.PY

```
$ GIT COMMIT -M "AI-ASSISTED  
ENHANCEMENTS: PERSISTENCE,  
COMPLETION, ERROR HANDLING"  
[AI-ENHANCEMENTS DEF5678] AI-  
ASSISTED ENHANCEMENTS:  
PERSISTENCE, COMPLETION, ERROR  
HANDLING 1 FILE CHANGED, 30  
INSERTIONS(+), 10 DELETIONS(-)
```

MERGED BACK TO MAIN.

```
$ GIT CHECKOUT MAIN  
SWITCHED TO BRANCH 'MAIN'  
YOUR BRANCH IS UP TO DATE WITH 'MAIN'.
```

```
$ GIT MERGE AI-ENHANCEMENTS  
UPDATING  
ABC1234..DEF5678  
FAST-FORWARD
```

TODO.PY | 40

+++++

----- 1 FILE

CHANGED, 20

INSERTIONS(+), 20

DELETIONS(-)

REFLECTION: GIT BRANCHING ALLOWED SAFE
TESTING OF AI CHANGES WITHOUT BREAKING
MAIN. MERGING INTEGRATED IMPROVEMENTS
CLEANLY, SIMULATING PAIR PROGRAMMING
COLLAB.

TASK 3: AI FOR COMMIT MESSAGE GENERATION
CHANGES MADE: ADDED PERSISTENCE WITH
JSON.

HUMAN-WRITTEN COMMIT: "ADD FILE SAVE"

ANOTHER EXAMPLE:

CHANGES: ADDED COMPLETION TOGGLE.

HUMAN: "ADD DONE FEATURE"

AI: "FEAT(TODO): INTRODUCE COMPLETION STATUS WITH TOGGLE FUNCTION AND VISUAL INDICATOR IN LIST VIEW"

AI MESSAGES ARE MORE DESCRIPTIVE, USE CONVENTIONAL COMMITS (FEAT:, FIX:), IMPROVE TRACEABILITY IN GIT LOG.

```
$ GIT LOG --ONELINE DEF5678 (HEAD ->
MAIN) FEAT(PERSISTENCE): ADD JSON
FILE SAVING/LOADING...
```

```
ABC1234 INITIAL AI-GENERATED TO-DO
LIST MANAGER
```

BETTER DOCS FOR TEAM REVIEWS.

TASK 4: PAIR PROGRAMMING SIMULATION

STUDENT A INITIAL CODE (SIMPLE CALCULATOR
INSTEAD OF TODO FOR VARIETY):

```
#  
  
CALC_INITIAL.PY  
  
BY STUDENT A  
  
DEF ADD(A, B):  
  
    RETURN A + B  
  
  
DEF SUB(A,  
  
B):  
  
    RETURN A - B  
  
  
PRINT(ADD(5, 3))  
  
# 8 PRINT(SUB(5,  
  
3)) # 2
```


STUDENT B WITH AI: SUGGESTED MENU, ERROR HANDLING, MUL/DIV.

IMPROVED:

```
#
```

```
CALC_IMPROVED.PY
```

```
DEF ADD(A, B):
```

```
    RETURN A + B DEF
```

```
SUB(A, B): RETURN
```

```
A - B DEF MUL(A,
```

```
B): RETURN A * B
```

```
DEF DIV(A, B):
```

```
    TRY:
```

```
        RETURN A / B
```

```
    EXCEPT
```

```
ZERODIVISIONERROR:
```

```
        RETURN "ERROR: DIVISION BY ZERO"
```

```
WHILE TRUE:  PRINT("1.ADD  
2.SUB 3.MUL 4.DIV 5.QUIT")  CH  
= INPUT()  IF CH == '5': BREAK  
A, B = MAP(FLOAT, INPUT("NUMS:  
").SPLIT())  IF CH == '1':  
PRINT(ADD(A,B))  ELIF CH ==  
'2': PRINT(SUB(A,B))  ELIF CH  
== '3': PRINT(MUL(A,B))  ELIF  
CH == '4': PRINT(DIV(A,B))
```

GIT WORKFLOW:

```
$ GIT INIT CALC-LAB  
$ GIT ADD CALC_INITIAL.PY  
$ GIT COMMIT -M "STUDENT A: BASIC  
ADD/SUB CALC"  
  
$ GIT CHECKOUT -B STUDENT-B-AI
```

```
$ MV CALC_IMPROVED.PY CALC.PY #  
RENAME AND REPLACE  
$ GIT ADD CALC.PY  
$ GIT RM CALC_INITIAL.PY  
$ GIT COMMIT -M "STUDENT B: AI-  
ENHANCED CALCULATOR WITH MENU,  
MUL/DIV, ERRORS"  
$ GIT CHECKOUT MAIN  
$ GIT MERGE STUDENT-B-AI  
AUTO-MERGING CALC.PY
```

COLLABORATION: STUDENT A PROVIDED BASE,
B+AI POLISHED IT. GIT TRACKED CONTRIBUTIONS
CLEARLY.